

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

12

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| A) | Describe the layers of the Earth's atmosphere and their impact on radio wave propagation. | Understand | 6 |
| B) | Define MUF, A high frequency radio link has to be established between two points at a distance of 2500 km on earth's surface. Consider ionospheric height to be 220 km and its critical frequency 4 MHz, Calculate the MUF for given path. | Evaluate | 6 |
| C) | Derive an expression for Friis Transmission Equation, and explain its significance on signal propagation in wireless communication systems? | Analyze | 6 |

Q.2 Solve Any Two of the following.

12

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| A) | Explore the various types of antennas and their applications in different communication scenarios. | Analyze | 6 |
| B) | Define and elaborate on terms such as radiation pattern, radiation power density, radiation intensity, directivity, and gain. | Remember | 6 |
| C) | Derive an expression for the fields radiated by infinitesimal dipole antenna. | Analyze | 6 |

Q. 3 Solve Any Two of the following.

12

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| A) | Provide a detailed explanation of a log periodic antenna along with its unique features, Also explain its different applications. | Understand | 6 |
| B) | Compare the characteristics of a binomial and Dolph Tchebyshev antenna array. Explain the design principles behind each type and how they affect the antenna's performance. | Analyze | 6 |
| C) | Discuss the principle of pattern multiplication in an N-element linear array. How does the radiation pattern evolve as the number of elements increases? | Understand | 6 |

Q.4 Solve Any Two of the following.

12

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|----|--|---------|---|
| A) | Explain the advantages of smart antennas in wireless communication systems. | Analyze | 6 |
| B) | Compare adaptive and fixed beamforming techniques along with the distinguishing features of each, and in what scenarios are they most effective? | Analyze | 6 |

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| C) | Explain different challenges associated with the implementation of smart antennas in communication systems. | Analyze | 6 |
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Q. 5	Solve Any Two of the following.		12
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| A) | Write a short note on (a) Rhombic Antenna and (b) V antenna. | Remember | 6 |
| B) | Provide a detailed explanation of the microstrip antenna's structure, and discuss the different methods used for feeding this type of antenna. | Understand | 6 |
| C) | Explain the structural characteristics and dimensions of antennas with parabolic reflectors. Discuss the features and applications that make antennas with parabolic reflectors effective for long-range communication. | Analyze | 6 |

***** End *****